

# POLITECNICO DI TORINO

Master's Degree in Data Science and Engineering



## Politecnico di Torino

Master's Degree Thesis

# Predictive Analytics of Climate Stress Impact on the Italian Mediterranean Buffalo: Physiological and Productive Responses

Supervisors

Prof. Luca BARBIERATO

Prof. Edoardo PATTI

Prof. Gianluca NEGLIA

Prof. Matteo SANTINELLO

Candidate

Seyed Mohammad SHEIKH AHMADI GANDAB

Academic Year 2025-2026



# Summary

# Acknowledgements

*To my parents and beloved.*



# Table of Contents

|                         |      |
|-------------------------|------|
| List of Tables          | VI   |
| List of Figures         | VII  |
| List of Listing         | VIII |
| List of Acronyms        | IX   |
| 1 Introduction          | 1    |
| 2 Enabling Technologies | 2    |
| 3 Literature Review     | 3    |
| 4 Methodology           | 4    |
| 5 Scenario              | 5    |
| 6 Experimental Results  | 6    |
| 7 Conclusions           | 7    |
| Bibliography            | 8    |

# List of Tables

# List of Figures



# List of Listings

# List of Acronyms

**IA** Interface Algorithm

# Chapter 1

## Introduction

Interface Algorithm (IA) Interface Algorithm IA

## Chapter 2

# Enabling Technologies

## Chapter 3

# Literature Review

In [1],

## Chapter 4

# Methodology

# Chapter 5

## Scenario

## Chapter 6

# Experimental Results



## Chapter 7

# Conclusions

# Bibliography

- [1] Mathias Uslar et al. «Applying the smart grid architecture model for designing and validating system-of-systems in the power and energy domain: A European perspective». In: *Energies* 12.2 (2019), p. 258 (cit. on p. 3).